

Marco Villasuso

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Summary

Aerospace engineering undergraduate at the University of Colorado Boulder (3.56 GPA, expected May 2028) with hands-on experience in propulsion, flight vehicle systems, CAD, FEA, CFD, and hardware integration. Led a team of ~10 people through design and iteration of a scale flight vehicle and supported a ~800 lbf Ethanol/LOX engine hot-fire. Experienced in engineering design reviews, hardware testing, troubleshooting, and cross-disciplinary systems integration.

Education

University of Colorado Boulder - BS Aerospace Engineering August 2024 - May 2028 (Expected)

- Maintained a GPA of 3.56 / 4.0, Dean's List for 3 of 4 semesters
 - Relevant courses: aerodynamics, structures, vehicle dynamics, systems & integration
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Projects / Work Experience / Leadership

Big Dumb Rockets (BDR)

August 2025 - Present

- **Team lead** for **Hopper Demo**, a scale demo designed to test flight vehicle systems including suspension, GNC, thrust vectoring, flight dynamics, and flight computers. Coordinated interfaces with co-leads across all BDR subteams for systems and their integration into the flight vehicle. Led a ~10 person team through trade studies, design, and reviews to coordinate a final design that met scale parameters and stayed within budget.
- **Team lead** for **Capacitive Fill**, a cryogenic tank level measurement system. Coordinated a team of 3 through design, manufacturing, and integration. Worked with machinists for integration into model tanks. Successfully measured tank fill level from empty to full with $\pm 5\%$ accuracy.
- Contributed structural and fluid analysis to **Regen Mk II**, a regeneratively cooled Ethanol/LOX engine targeting long duration, throttleable and gimbaled burns.
- Performed hardware integration and acceptance testing for **Regen Mk I**, including component inspection, assembly verification, and post-test teardown analysis. Assisted with gasket cutting, injector manifold machining, and LOX-cleaning plumbing lines before integration with Regen Mk I.
- Participated in propulsion testing & on-site in-the-moment troubleshooting for Regen Mk I, supporting a successful ~800 lbf hot-fire.

RC Jet (Solo)

July 2026 - August 2026

- Independently designed, simulated, built, and flew a ducted-fan powered remote controlled jet. Designed in OnShape, ran FEA simulations for wing loading in PrepoMax, and performed flow analysis in flow5, estimating a top speed of ~50 mph. Fabricated via 3D printing through 3 design iterations. Flew a successful powered flight for 30 seconds before landing safely in grass.

CU Book Store

August 2024 - Present

- **Student Lead for Technology** (December 2024 - Present)
 - Managed and supported technical systems including **3D printers, laser cutters, computers, tablets, kiosks, and CAD-related equipment**.
 - Troubleshoot hardware and software issues and coordinated repairs and deployment of technical equipment.
 - Tested and deployed a rental-system implementation and trained staff on new equipment and procedures.
 - Conducted employee training and performance evaluations while coordinating technical tasks across the team.
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Technical Skills

- **CAD:** SOLIDWORKS, OnShape
- **Manufacturing:** 3D printing (PLA, ABS, ASA, SLA), laser cutting, 2-axis CNC, MIG welding
- **FEA:** PrepoMax, SOLIDWORKS Simulation
- **CFD:** Ansys Fluent, OpenFOAM, flow5
- **Systems:** Systems Integration, Flight Hardware, Propulsion Systems, Cryogenic Systems
- **Testing & Operations:** Engine Hot-Fires, Hardware Acceptance Testing, LOX Handling, Troubleshooting
- **Programming:** C & C++, Python, MATLAB, Lua